

Sumin Lee

cirtuare@snu.ac.kr

www.linkedin.com/in/cirtuare • <https://github.com/cirtuare>

RESEARCH INTERESTS

My research interests lie in **human-AI interaction**, with a growing curiosity toward how humans and AI collaborate — how multimodal engagement with AI shapes not only the outcomes of that collaboration but also people's understanding and wellbeing. At the core is an interest in how AI can improve quality of life, not only in high-impact applications but in the incremental ways that shape everyday experience.

EDUCATION

Seoul National University

2023 – 2027 (Expected)

B.S. in Civil, Urban & Environmental Engineering, B.A. in Information Science and Culture

RESEARCH EXPERIENCE

IMSI LAB, SNU

Feb 2026 – Present

Research Intern (Advisor: Prof. Nam-Joon Kim)

- Led dataset curation, aggregating 10 dental datasets into a single standardized benchmark with visualization pipeline
- Conducted ablation studies training deep learning models across multiple ongoing medical imaging projects

PUBLICATIONS

C: conference, P: preprint, *: equal contribution, †: corresponding author

- p3 **I'm Fine, But My Voice Isn't: Cross-Modal Affective Dissonance Detection for Reflective Journaling**
Sumin Lee[†]
Under review
- c2 **3D Tactile Display for Non-visual Color Perception**
Sumin Lee^{*†}, Yehyeon Park^{*}
Proceedings of HCI Korea 2026
- p2 **NEMESIS: Noise-suppressed Efficient MAE with Enhanced Superpatch Integration Strategy**
Kyeonghun Kim, Hyeonseok Jung, Youngung Han, Hyunsu Go, Eunseob Choi, Seongbin Park, Junsu Lim, Jiwon Yang, Sumin Lee, Insung Hwang, Ken Ying-Kai Liao, Nam-Joon Kim[†]
Under review
- p1 **MATHENA: Mamba-based Architectural Tooth Hierarchical Estimator and Holistic Evaluation Network for Anatomy**
Kyeonghun Kim, Jaehyung Park, Youngung Han, Anna Jung, Seongbin Park, Sumin Lee, Jiwon Yang, Jiyeon Han, Subeen Lee, Junsu Lim, Hyunsu Go, Eunseob Choi, Hyeonseok Jung, Soo Yong Kim, Woo Kyoung Jeong, Won Jae Lee, Pa Hong, Hyuk-Jae Lee, Ken Ying-Kai Liao, Nam-Joon Kim[†]
Under review
- c1 **Memory and Spatial Patterns in Long-Term Time Series of River Discharge and Water Quality Parameters Using Fractal Theory**
Eunpyo Lee, Jungsoo Woo, Sumin Lee, Soyun Ihn, Soohyun Yang[†]
KSCE 2024 Convention

HONORS & AWARDS

AI-SW Maestro Trainee

Apr 2026 – Present

Nationwide Software Training Program with scholarship approximately US \$6,800 (Ministry of Science and ICT)

Excellence Award, SNU Social Responsibility Contest

Nov 2025

SELECTED PROJECTS

Color 3D: 3D Tactile Color Display for Visual Accessibility

Jul 2025 – Dec 2025

Software Developer & Team Lead | HCI, Assistive Technology

- International Conference on Appropriate Technology (ICAT) 2025 Speaker
- Developed full-stack tactile display system enabling visually impaired users to independently perceive colors through haptic feedback
- Implemented an iOS application featuring multimodal interaction via color-height mapping algorithms, voice-guided interfaces, and on-device AI descriptions
- Engineered ESP32 firmware for 24-motor control system with Bluetooth communication protocol for real-time tactile rendering

River Discharge & Water Quality Time-Series Data Analysis

Mar 2024 – Nov 2024

2024 WISER Engineering Research Team G6 Member | Hydrology

- Developed a comprehensive Python data pipeline that efficiently processed 10-year multi-resolution time series data across 14 water quality parameters from Korea's major river basins
- Implemented signal processing techniques (FFT, power spectrum analysis) using NumPy and SciPy to identify memory patterns and fractal characteristics in environmental data, creating various log-log visualizations

SNU Club App “Allclear”

Sep 2025 – Present

Mobile Developer | React Native

- Migrated iOS app to a new Apple Developer account, resolving code signing and deployment issues
- Led Admin Feature TF, defining requirements and driving decisions between design and backend teams

SNU Dining App “Siksha”

Mar 2024 – Present

iOS Developer

- Maintained and improved SwiftUI app serving 10,000+ users, refactoring to clean architecture and implementing various features
- Diagnosed and resolved App Store rejection by revising privacy policy for anonymous content

Restaurant Finder App “Acon”

Jan 2025 – Oct 2025

iOS Lead Developer

- Spearheaded iOS development by proposing and architecting the app structure; managed App Store release end-to-end and led coordination across product, design, android, and backend teams to align priorities and resolve blockers
- Implemented memory-optimized custom photo album selection using Apple's Photos framework
- Designed extensible LocationManager with CoreLocation framework and Multicast delegate pattern and integrated it with Naver Map API, creating a location verification system
- Proposed and implemented Amplitude analytics via a custom AmplitudeManager class for user behavior tracking

Date Course Sharing App, “Dateroad”

Jul 2024 – Mar 2025

iOS Developer

- Developed complex UI components and engineered reusable custom alert controller to streamline use across multiple app sections
- Configured Google AdMob integration and implemented rewarded Google Video Ads

TEACHING

Computing Essentials

Spring 2026

Seoul National University F37.204(003) Tutor (Instructor: Prof. Yamada Akihiko)

- Supported 30 students with Python programming and debugging

LEADERSHIP

Wafflestudio (Seoul National University Web/App Development Club)

Aug 2024 – Aug 2025

Executive Committee, Project Team Leader

- Led 400+ members in web/app programming development, managing communications and driving engagement via Slack
- Organized alum homecoming event, bringing together members across 20 years of club history to create meaningful connections
- Served as the primary coordinator for collaboration with Seoul National University Festival Committee, synthesizing discussion points

TECHNICAL SKILLS & LANGUAGES

- Programming: Python, Swift, TypeScript, React Native
- Design & Prototyping: Figma, Cinema 4D, Adobe Premiere Pro, Illustrator
- Languages: English (Fluent - iBT TOEFL 109), Korean (Native)